

Pathogenwatch tutorial

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A short introduction

Pathogenwatch is a database of high quality genomes and metadata for a number of species, such as *Klebsiella pneumoniae*, *Staphylococcus aureus*, *Candida auris*, *Salmonella*, *Streptococcus pneumoniae*, SARS-CoV-2 and others, with robust bioinformatics capability for virulence, antimicrobial resistance, and finding related isolates. Pathogenwatch contains published genomes, and manually-curated metadata from the literature, including strain names; place, date and source of isolation; and phenotypic data for biotype, serogroup, serotype and antimicrobial resistance. Pathogenwatch is aimed for use by clinicians and public health workers studying outbreaks, as well as for research biologists investigating genomic variation and evolution.

Required internet browsers: Google Chrome or Mozilla Firefox

The aim of this tutorial is to give an overview of the features of Pathogenwatch.

- How to upload genomes to Pathogenwatch
- How to create a collection
- How to view the assembly statistics, the AMR determinants and sequence types of the uploaded genomes
- How to view the genome report.
- How to view clusters

The input files used in this tutorial can be found at <https://github.com/cgps-group/cluj-training/>

What would you like to upload?

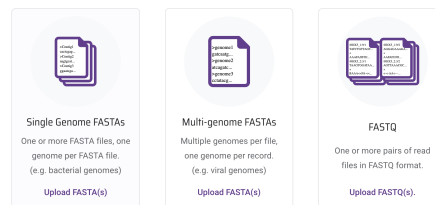


Figure 1: Go to pathogen.watch and log in. Go to the upload page, and then click on the “Single genomes fastas”

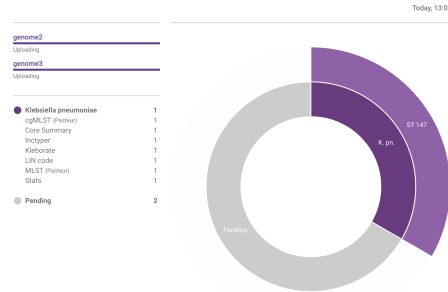


Figure 2: Drag and drop the genome assemblies from your computer to the browser window.

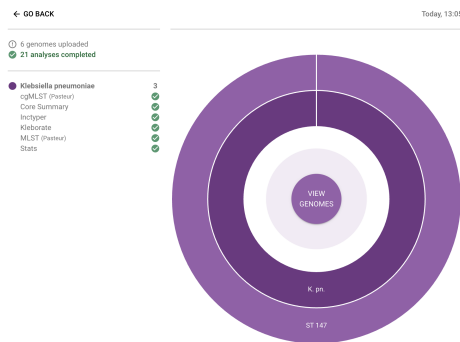


Figure 3: Wait until genomes are uploaded and all analytics are run on them. Then click on the “View Genomes button.”

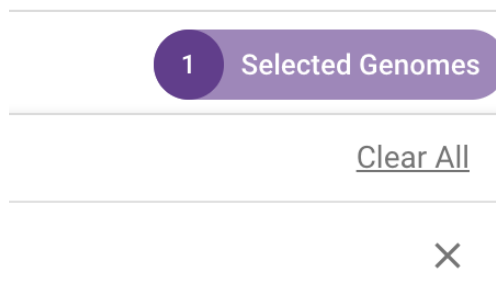


Figure 4: In the right hand side of the screen, check how many genomes are currently selected. In this example, there is one genome that has been selected in a previous analysis, which should not be part of the current analysis. Press the “Clear all” button to deselect this genome.

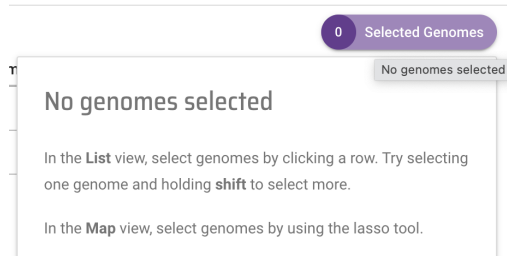


Figure 5: The “Selected Genomes” panel should show “0”.

Viewing 3 of 321,939 genomes

☒ Name	Organism
☒ genome1	<i>Klebsiella pneumoniae</i>
☒ genome2	<i>Klebsiella pneumoniae</i>
☒ genome3	<i>Klebsiella pneumoniae</i>

Figure 6: Make sure you select all genomes listed

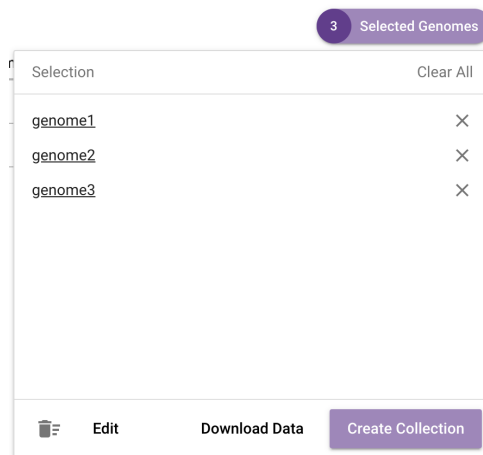


Figure 7: Go to the Selected genomes panel and press “Create collection”

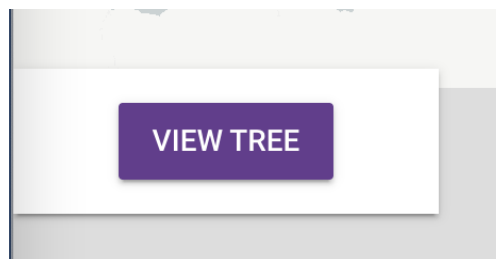


Figure 8: After the collection is begin created, press on “View tree”

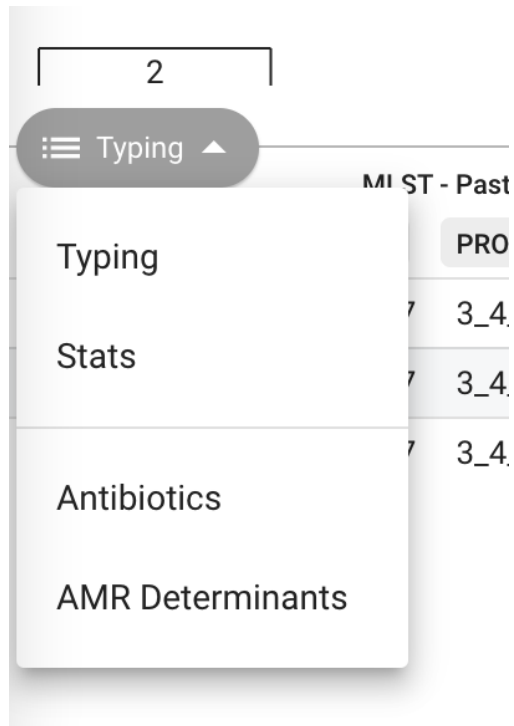


Figure 9: Explore the analytics run on the three uploaded genomes, such as typing, stats or Kleborate

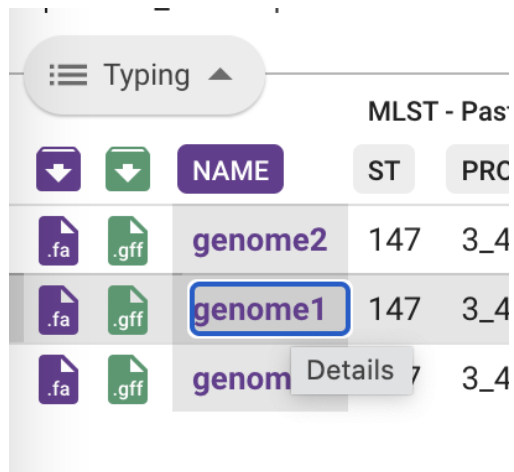


Figure 10: Click on one of the genome names.

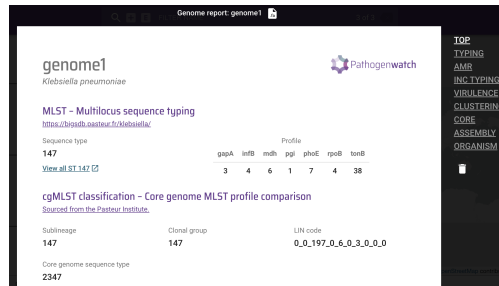


Figure 11: The genome report will be reported. Explore the data presented there.

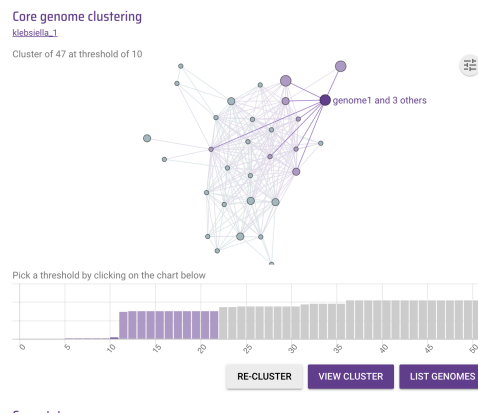


Figure 12: Explore the core genome clustering and press on “View cluster”

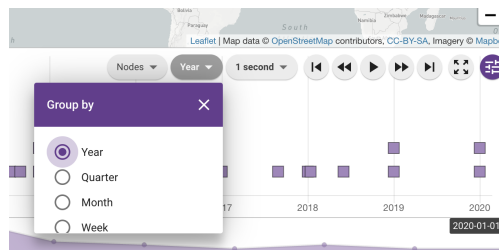


Figure 13: Adjust the timeline, so that it shows data by year.

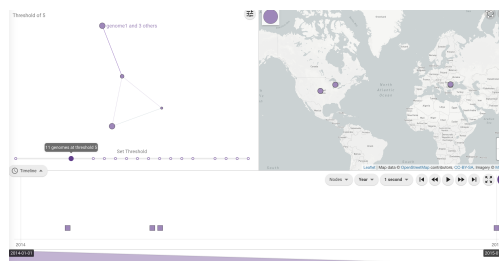


Figure 14: Lower the clustering threshold and check which are the closest genomes to the ones you have uploaded. What could genetic relatedness imply?